Current results for XES3G5M-sub-small

theta_m=0.8, theta_f=0.8, rec=50; remove training exfr0.00 edges

Train edges: mastery 279,895; forgetting 694,068; recommendation 36,350

N = 5, 10, ..., 50; Ep thresholds = 0.4, 0.5, 0.6, 0.7; planned seeds = 2024, 2025, 2026.

Cell = mean across available seed-level means +/- sample SD; a single result is shown without SD.

Bold = best; underline = second best among available models, ranked on unrounded means.

Cross-model rankings are provisional where seed coverage differs. CF baselines are deterministic.

Completed results included in this snapshot

Model	Seeds / evaluation	Section
2CKG4ER	2024,2025,2026	Ablation
No-mastery	2024,2025,2026	Ablation
No-forgetting	2024,2025,2026	Ablation
No-student-feature	2024,2025,2026	Ablation
No-exercise-feature	2024,2025,2026	Ablation
ID-only	2024,2025,2026	Ablation
EB-CF	deterministic	Comparison
SB-CF	deterministic	Comparison
CBF	deterministic	Comparison
TransE	2024,2025	Comparison
TransE-adv	-	Comparison
RotatE	-	Comparison
DistMult	-	Comparison
ComplEx	-	Comparison

Source: C:\Users\21709\Downloads\XES_current_results_20261010_095340.zip

Ablation | XES3G5M-sub-small

Mean +/- SD across available seeds; one result has no SD. Missing models are marked '-'.

Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-only
5	**0.879±0.003**	0.802±0.003	0.878±0.019	0.853±0.006	0.872±0.002	0.862±0.011
10	0.876±0.003	0.806±0.011	**0.880±0.010**	0.853±0.004	0.872±0.003	0.861±0.009
15	0.873±0.004	0.812±0.002	**0.878±0.009**	0.852±0.004	0.871±0.003	0.860±0.008
20	0.873±0.003	0.818±0.006	**0.877±0.009**	0.852±0.004	0.871±0.003	0.858±0.007
25	0.873±0.003	0.825±0.004	**0.877±0.009**	0.851±0.004	0.871±0.002	0.858±0.006
30	0.872±0.002	0.829±0.002	**0.877±0.008**	0.851±0.003	0.871±0.002	0.857±0.005
35	0.871±0.002	0.832±0.001	**0.877±0.008**	0.851±0.003	0.870±0.002	0.857±0.004
40	0.871±0.002	0.837±0.001	**0.876±0.007**	0.851±0.003	0.870±0.002	0.856±0.004
45	0.870±0.003	0.842±0.000	**0.876±0.006**	0.851±0.003	0.869±0.002	0.856±0.004
50	0.870±0.003	0.845±0.000	**0.876±0.005**	0.850±0.003	0.869±0.002	0.855±0.003

Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-only
5	**1.000±0.000**	1.000±0.000	0.998±0.001	0.999±0.001	1.000±0.000	1.000±0.000
10	**1.000±0.000**	1.000±0.000	0.997±0.001	0.999±0.001	1.000±0.000	1.000±0.000
15	1.000±0.000	**1.000±0.000**	0.997±0.002	0.999±0.001	1.000±0.000	1.000±0.000
20	1.000±0.000	**1.000±0.000**	0.996±0.002	0.999±0.000	1.000±0.000	1.000±0.000
25	1.000±0.000	**1.000±0.000**	0.996±0.002	1.000±0.000	1.000±0.000	1.000±0.000
30	1.000±0.000	**1.000±0.000**	0.994±0.002	0.999±0.000	1.000±0.000	1.000±0.000
35	1.000±0.000	**1.000±0.000**	0.994±0.001	1.000±0.000	1.000±0.000	1.000±0.000
40	1.000±0.000	**1.000±0.000**	0.993±0.002	1.000±0.000	1.000±0.000	1.000±0.000
45	1.000±0.000	**1.000±0.000**	0.992±0.003	1.000±0.000	1.000±0.000	1.000±0.000
50	1.000±0.000	1.000±0.000	0.992±0.003	1.000±0.000	1.000±0.000	**1.000±0.000**

Ablation | XES3G5M-sub-small

Mean +/- SD across available seeds; one result has no SD. Missing models are marked '-'.

Ep, response threshold 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-only
5	0.143±0.045	**0.243±0.018**	0.043±0.059	0.194±0.016	0.158±0.038	0.212±0.033
10	0.234±0.065	0.249±0.128	0.032±0.099	0.275±0.018	0.264±0.014	**0.285±0.031**
15	0.297±0.042	0.286±0.034	0.034±0.098	0.327±0.009	0.299±0.010	**0.330±0.027**
20	0.338±0.022	0.339±0.052	0.035±0.118	0.347±0.011	**0.355±0.024**	0.353±0.021
25	0.353±0.030	**0.436±0.007**	0.056±0.136	0.361±0.004	0.339±0.034	0.369±0.009
30	0.363±0.017	**0.452±0.043**	0.104±0.116	0.374±0.003	0.347±0.010	0.379±0.009
35	0.378±0.033	**0.513±0.029**	0.154±0.077	0.376±0.005	0.374±0.015	0.384±0.003
40	0.378±0.017	**0.541±0.023**	0.206±0.044	0.378±0.005	0.379±0.015	0.388±0.005
45	0.394±0.012	**0.500±0.045**	0.214±0.045	0.378±0.001	0.389±0.038	0.383±0.004
50	0.391±0.007	**0.465±0.075**	0.237±0.042	0.372±0.005	0.379±0.027	0.386±0.009

Ep, response threshold 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-only
5	0.138±0.045	**0.238±0.020**	0.040±0.057	0.189±0.015	0.154±0.039	0.208±0.033
10	0.230±0.064	0.246±0.128	0.021±0.090	0.271±0.018	0.258±0.017	**0.283±0.032**
15	0.292±0.040	0.283±0.034	0.024±0.092	0.322±0.010	0.294±0.011	**0.327±0.028**
20	0.333±0.020	0.336±0.053	0.022±0.108	0.342±0.011	**0.351±0.023**	0.351±0.022
25	0.349±0.028	**0.434±0.008**	0.043±0.128	0.356±0.003	0.336±0.034	0.367±0.010
30	0.359±0.015	**0.450±0.043**	0.093±0.112	0.370±0.004	0.344±0.010	0.377±0.009
35	0.374±0.032	**0.513±0.029**	0.134±0.072	0.372±0.005	0.371±0.015	0.383±0.002
40	0.375±0.014	**0.541±0.023**	0.186±0.041	0.375±0.005	0.377±0.015	0.387±0.005
45	0.391±0.009	**0.500±0.045**	0.195±0.038	0.376±0.002	0.387±0.037	0.382±0.004
50	0.389±0.004	**0.465±0.075**	0.216±0.035	0.370±0.004	0.378±0.026	0.386±0.009

Ablation | XES3G5M-sub-small

Mean +/- SD across available seeds; one result has no SD. Missing models are marked '-'.

Ep, response threshold 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-only
5	0.127±0.043	**0.229±0.022**	0.037±0.057	0.178±0.013	0.137±0.040	0.200±0.030
10	0.216±0.061	0.238±0.127	0.005±0.078	0.256±0.020	0.232±0.025	**0.272±0.032**
15	0.277±0.033	0.277±0.035	0.001±0.079	0.306±0.012	0.268±0.016	**0.315±0.032**
20	0.317±0.013	0.331±0.053	-0.014±0.101	0.324±0.014	0.324±0.025	**0.340±0.028**
25	0.332±0.020	**0.429±0.008**	0.003±0.122	0.339±0.004	0.309±0.039	0.357±0.017
30	0.342±0.011	**0.447±0.043**	0.049±0.112	0.354±0.004	0.317±0.015	0.367±0.016
35	0.359±0.026	**0.511±0.029**	0.092±0.076	0.357±0.003	0.346±0.014	0.375±0.006
40	0.359±0.006	**0.540±0.023**	0.133±0.047	0.360±0.008	0.354±0.016	0.380±0.003
45	0.376±0.008	**0.500±0.045**	0.141±0.032	0.362±0.005	0.365±0.030	0.375±0.008
50	0.374±0.008	**0.465±0.075**	0.155±0.034	0.358±0.001	0.357±0.018	0.380±0.004

Ep, response threshold 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-only
5	0.091±0.045	**0.211±0.025**	0.033±0.056	0.157±0.010	0.100±0.047	0.177±0.029
10	0.164±0.061	0.218±0.120	-0.024±0.059	0.223±0.020	0.165±0.040	**0.241±0.031**
15	0.208±0.023	0.256±0.030	-0.033±0.055	0.265±0.014	0.192±0.026	**0.282±0.033**
20	0.243±0.006	**0.307±0.048**	-0.063±0.082	0.277±0.021	0.241±0.040	0.304±0.032
25	0.251±0.011	**0.399±0.010**	-0.059±0.098	0.289±0.016	0.215±0.057	0.317±0.026
30	0.262±0.010	**0.417±0.044**	-0.028±0.095	0.301±0.014	0.221±0.031	0.325±0.026
35	0.277±0.019	**0.481±0.031**	0.005±0.070	0.304±0.019	0.247±0.017	0.331±0.020
40	0.276±0.005	**0.515±0.023**	0.031±0.042	0.308±0.016	0.256±0.019	0.335±0.020
45	0.293±0.011	**0.480±0.045**	0.025±0.056	0.309±0.020	0.267±0.017	0.331±0.025
50	0.291±0.019	**0.448±0.076**	0.027±0.055	0.303±0.024	0.258±0.020	0.334±0.019

Comparison | XES3G5M-sub-small

Mean +/- SD across available seeds; one result has no SD. Missing models are marked '-'.

Adaptability (Ada)

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	0.821	0.834	0.549	0.846±0.000	-	-	-	-	**0.879±0.003**
10	0.811	0.840	0.577	0.850±0.000	-	-	-	-	**0.876±0.003**
15	0.811	0.844	0.594	0.851±0.000	-	-	-	-	**0.873±0.004**
20	0.813	0.846	0.604	0.850±0.001	-	-	-	-	**0.873±0.003**
25	0.815	0.846	0.613	0.848±0.002	-	-	-	-	**0.873±0.003**
30	0.817	0.847	0.621	0.846±0.002	-	-	-	-	**0.872±0.002**
35	0.818	0.845	0.628	0.845±0.002	-	-	-	-	**0.871±0.002**
40	0.820	0.842	0.633	0.845±0.001	-	-	-	-	**0.871±0.002**
45	0.821	0.840	0.638	0.845±0.001	-	-	-	-	**0.870±0.003**
50	0.822	0.839	0.642	0.845±0.001	-	-	-	-	**0.870±0.003**

Novelty (NOV)

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	0.998	1.000	**1.000**	**1.000±0.000**	-	-	-	-	**1.000±0.000**
10	0.998	1.000	**1.000**	**1.000±0.000**	-	-	-	-	1.000±0.000
15	0.999	1.000	**1.000**	**1.000±0.000**	-	-	-	-	1.000±0.000
20	0.999	0.999	1.000	**1.000±0.000**	-	-	-	-	1.000±0.000
25	0.998	0.999	1.000	**1.000±0.000**	-	-	-	-	1.000±0.000
30	0.996	0.998	1.000	**1.000±0.000**	-	-	-	-	1.000±0.000
35	0.994	0.997	1.000	**1.000±0.000**	-	-	-	-	1.000±0.000
40	0.993	0.998	1.000	**1.000±0.000**	-	-	-	-	1.000±0.000
45	0.993	0.998	1.000	**1.000±0.000**	-	-	-	-	1.000±0.000
50	0.992	0.998	1.000	**1.000±0.000**	-	-	-	-	1.000±0.000

Comparison | XES3G5M-sub-small

Mean +/- SD across available seeds; one result has no SD. Missing models are marked '-'.

Ep, response threshold 0.4

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	-0.021	0.182	0.117	**0.207±0.011**	-	-	-	-	0.143±0.045
10	-0.023	0.182	0.191	**0.306±0.018**	-	-	-	-	0.234±0.065
15	-0.054	0.239	0.244	**0.355±0.004**	-	-	-	-	0.297±0.042
20	-0.051	0.210	0.284	**0.385±0.006**	-	-	-	-	0.338±0.022
25	-0.029	0.202	0.293	**0.378±0.000**	-	-	-	-	0.353±0.030
30	-0.024	0.206	0.308	**0.381±0.003**	-	-	-	-	0.363±0.017
35	-0.049	0.192	0.311	0.378±0.003	-	-	-	-	**0.378±0.033**
40	-0.050	0.196	0.312	0.375±0.003	-	-	-	-	**0.378±0.017**
45	-0.055	0.218	0.318	0.364±0.004	-	-	-	-	**0.394±0.012**
50	-0.038	0.253	0.325	0.360±0.010	-	-	-	-	**0.391±0.007**

Ep, response threshold 0.5

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	-0.021	0.175	0.079	**0.204±0.012**	-	-	-	-	0.138±0.045
10	-0.023	0.174	0.151	**0.301±0.021**	-	-	-	-	0.230±0.064
15	-0.054	0.232	0.208	**0.350±0.006**	-	-	-	-	0.292±0.040
20	-0.052	0.201	0.254	**0.381±0.004**	-	-	-	-	0.333±0.020
25	-0.030	0.193	0.268	**0.374±0.001**	-	-	-	-	0.349±0.028
30	-0.024	0.195	0.291	**0.379±0.002**	-	-	-	-	0.359±0.015
35	-0.049	0.181	0.295	**0.376±0.004**	-	-	-	-	0.374±0.032
40	-0.051	0.186	0.299	0.374±0.002	-	-	-	-	**0.375±0.014**
45	-0.055	0.210	0.308	0.364±0.003	-	-	-	-	**0.391±0.009**
50	-0.038	0.247	0.320	0.360±0.010	-	-	-	-	**0.389±0.004**

Comparison | XES3G5M-sub-small

Mean +/- SD across available seeds; one result has no SD. Missing models are marked '-'.

Ep, response threshold 0.6

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	-0.021	0.166	0.050	**0.192±0.015**	-	-	-	-	0.127±0.043
10	-0.023	0.163	0.114	**0.286±0.022**	-	-	-	-	0.216±0.061
15	-0.054	0.220	0.159	**0.335±0.010**	-	-	-	-	0.277±0.033
20	-0.052	0.186	0.200	**0.365±0.001**	-	-	-	-	0.317±0.013
25	-0.030	0.171	0.219	**0.359±0.006**	-	-	-	-	0.332±0.020
30	-0.024	0.165	0.246	**0.365±0.003**	-	-	-	-	0.342±0.011
35	-0.049	0.147	0.254	**0.363±0.007**	-	-	-	-	0.359±0.026
40	-0.051	0.151	0.254	**0.361±0.000**	-	-	-	-	0.359±0.006
45	-0.055	0.178	0.270	0.353±0.002	-	-	-	-	**0.376±0.008**
50	-0.038	0.216	0.284	0.351±0.008	-	-	-	-	**0.374±0.008**

Ep, response threshold 0.7

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	-0.022	0.132	-0.010	**0.174±0.008**	-	-	-	-	0.091±0.045
10	-0.024	0.127	0.032	**0.266±0.021**	-	-	-	-	0.164±0.061
15	-0.055	0.184	0.070	**0.310±0.011**	-	-	-	-	0.208±0.023
20	-0.053	0.141	0.105	**0.338±0.004**	-	-	-	-	0.243±0.006
25	-0.031	0.117	0.106	**0.330±0.010**	-	-	-	-	0.251±0.011
30	-0.025	0.093	0.127	**0.337±0.009**	-	-	-	-	0.262±0.010
35	-0.053	0.041	0.132	**0.334±0.016**	-	-	-	-	0.277±0.019
40	-0.056	0.048	0.131	**0.333±0.010**	-	-	-	-	0.276±0.005
45	-0.063	0.054	0.150	**0.322±0.010**	-	-	-	-	0.293±0.011
50	-0.045	0.099	0.177	**0.319±0.001**	-	-	-	-	0.291±0.019